

The Solvability of Chess

An Analysis of the Elon Musk–Chess.com Exchange

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Abstract

In September 2026, Elon Musk and Chess.com engaged in a public exchange on X concerning the computational complexity and eventual solvability of chess. What initially appeared to be a brief social-media argument developed into a discussion involving several distinct mathematical concepts: the number of possible chess games, the number of legal chess positions, computational tractability, and the meaning of the word “solved.”

This essay examines the exchange as both a technical disagreement and a case study in how complex mathematical ideas are compressed into short-form social-media arguments. Particular attention is given to the distinction between the size of the chess game tree and the number of reachable legal positions. The analysis also considers whether “solving chess” necessarily requires explicitly enumerating every possible game or position.

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1 Introduction

Chess occupies an unusual position among games. Its rules are simple enough to be explained in a few pages, yet the number of possible positions and game sequences is so large that exhaustive analysis is far beyond the practical capabilities of present-day computers.

Modern chess engines have nevertheless become extraordinarily strong. Systems based on algorithms such as alpha–beta search, Monte Carlo methods, evaluation functions, neural networks, and enormous computational resources can already play chess at a level far beyond ordinary human players.

This creates an interesting question:

Is chess merely extremely difficult to solve, or is it ultimately solvable in a practical sense?

The distinction became the subject of a public exchange between Elon Musk and the Chess.com account on X in September 2026.

Musk argued that the number of genuinely useful moves in chess is small and predicted that chess would eventually be “fully solved” in a manner analogous to checkers. Chess.com’s response emphasized the enormous number of possible games.

At first glance, the disagreement appears to concern a simple question of arithmetic. In reality, the exchange mixes several different questions about the structure of chess.

This essay separates those questions.

2 The Exchange

2.1 Musk’s Initial Claim

On September 3, 2026, Musk argued on X that the number of moves in chess that are not effectively useless is very small and that chess would eventually be fully solved.

The important part of his argument was not merely that computers are already stronger than humans. Rather, it was the extrapolation from present computational superiority toward eventual mathematical solvability.

That distinction matters.

A game can be computationally manageable for a machine in many individual positions while remaining enormously difficult to solve globally.

2.2 Chess.com’s Response

Chess.com’s official X account responded with the deliberately terse phrase:

“Skill issue.”

The exchange subsequently became more technical.

Chess.com emphasized the enormous number of possible chess games, arguing that the game tree is vastly larger than the number of atoms in the observable universe.

Musk objected to this comparison and redirected the discussion toward the number of legal chess positions.

This was the point at which the argument became substantially more interesting.

3 Two Different Numbers

The central mathematical confusion in the exchange comes from treating two related quantities as though they were interchangeable.

They are not.

3.1 Possible Games

A chess game is a sequence of positions produced by successive legal moves.

Even if the number of legal positions were relatively manageable, the number of possible sequences through those positions could be enormous.

A rough historical estimate commonly associated with Claude Shannon places the number of possible chess games around

$$10^{120}.$$

This figure is often called the “Shannon number.”

It should not be interpreted as an exact enumeration of every possible game. It is an order-of-magnitude estimate illustrating the enormous branching factor of chess.

3.2 Legal Positions

A different quantity is the number of legal positions that can occur on a chessboard.

Estimates for the number of legal positions are dramatically smaller than the estimated number of complete game sequences.

This distinction supports an important part of Musk’s objection.

If the objective were literally to store every possible game sequence, the task would be vastly larger than storing every position.

However, that does not mean that storing every legal position would automatically solve chess.

4 What Does “Solved” Mean?

The word “solved” is doing a surprising amount of work in the debate.

There are several different meanings.

4.1 Weakly Solved

A game is weakly solved if the theoretical result from the initial position is known under perfect play.

For example, if perfect play from the initial position necessarily produces a draw, the game is weakly solved if that fact has been mathematically established.

4.2 Strongly Solved

A game is strongly solved when the theoretically optimal result can be determined from every reachable position.

This is a substantially stronger requirement.

For chess, strong solution would involve determining the optimal outcome of arbitrary legal positions, not merely determining the result from the initial board.

4.3 Practical Solving

There is also a third concept that is often implicit in discussions about artificial intelligence:

A system may be capable of playing essentially perfect chess without possessing a complete mathematical solution to chess.

This distinction is crucial.

A sufficiently powerful engine could potentially become practically unbeatable without humanity ever possessing a compact table containing the theoretically optimal continuation from every possible position.

5 Game Trees and Compression

Musk's most interesting argument concerns the possibility that a future artificial intelligence system could discover a much more efficient representation of the problem.

Suppose, for illustration, that a game contains a set of states

$$S = \{s_1, s_2, \dots, s_n\}.$$

A naive approach might attempt to examine every possible sequence of moves.

But many different sequences can eventually reach the same position. Consequently, analyzing every sequence independently would duplicate work.

This is one reason that modern chess engines do not simply enumerate every possible game.

They exploit the structure of the search space.

For example, if several different move sequences lead to the same position s , the value of s can potentially be reused:

$$V(s) = f(s).$$

Instead of calculating the consequences of every path independently, an algorithm can calculate the value of the position and reuse that result.

This is conceptually similar to dynamic programming.

Therefore:

Number of possible games $\neq$ computational work required to solve the game.

The difference between those quantities is one of the most important points obscured by the original social-media argument.

6 Why the Atom Comparison Is Both Useful and Limited

Chess.com's comparison between possible games and atoms in the observable universe is rhetorically powerful because it gives a human-scale intuition for an otherwise incomprehensible number.

However, it is not by itself an argument that chess cannot be solved.

Consider the following simplified analogy.

Suppose a problem has

$$10^{120}$$

possible paths but only

$$10^{45}$$

meaningfully distinct states.

The number of paths may be astronomically large while the underlying state space is considerably smaller.

Furthermore, a mathematical solution does not necessarily require explicitly listing every path.

A sufficiently compact theorem, algorithm, or representation could in principle describe the behavior of an enormous number of states.

This is why the statement

“There are more possible games than atoms”

should not be treated as equivalent to

“Chess cannot be solved.”

The former concerns the size of a combinatorial space.

The latter is a claim about the existence and computational tractability of a solution.

7 The Role of Artificial Intelligence

The development of increasingly powerful chess engines demonstrates that enormous search spaces can be attacked without literally enumerating every possibility.

Modern systems combine search, evaluation, learned representations, and extensive prior computation.

This changes the practical question.

Instead of asking:

“Can a computer examine every possible game?”

one can ask:

“Can a computer construct a sufficiently compressed representation of chess that determines optimal play?”

Those are different problems.

The second question is much closer to the argument Musk was making.

Whether an artificial intelligence system could eventually discover such a representation is an open technological and mathematical question rather than something established by the number of legal positions alone.

8 The Rhetoric of the X Exchange

The exchange is also interesting because the technical disagreement was embedded inside the communication style of X.

Chess.com’s initial “skill issue” response was intentionally provocative rather than explanatory.

Musk responded by defending the underlying proposition and eventually criticized the use of the observable-universe comparison.

The discussion subsequently became partly about the identity and status of the person operating the Chess.com account. Musk referred to the account operator as an “intern,” while the account operator responded that they were a full-time employee.

This illustrates a recurring feature of technical arguments on social media: the discussion can rapidly move between several levels.

1. The original technical proposition.
2. The evidence supporting that proposition.
3. The credibility of the participants.
4. The rhetorical framing of the disagreement.
5. The audience’s interpretation of the exchange.

The result is entertaining, but it also makes careful technical analysis more difficult.

9 Where the Arguments Intersect

The most interesting conclusion is that the two sides were not necessarily answering exactly the same question.

Concept	What it describes	Relevance
Game tree	Possible sequences of moves	Measures the enormous branching structure of chess
Legal positions	Distinct reachable board states	Measures the underlying state space
Perfect play	Optimal decisions from positions	Describes what a solution would need to determine
Practical engine strength	Ability to choose strong moves	Does not necessarily constitute a mathematical solution
Strong solution	Optimal result from arbitrary legal positions	A much stronger requirement than merely playing extremely well

Table 1: Different concepts involved in the solvability question.

Chess.com’s argument demonstrates why chess is extraordinarily difficult.

Musk’s response highlights why the raw number of possible games is not, by itself, sufficient to establish impossibility.

The deeper question lies between them:

How much of the enormous chess state space can be compressed into a representation from which optimal play can be derived?

That is a considerably more interesting problem than the original social-media exchange suggests.

10 Conclusion

The Elon Musk–Chess.com exchange began as a short argument about whether chess would eventually be solved, but it exposed a genuine conceptual distinction.

The number of possible chess games and the number of legal chess positions are not the same quantity. More importantly, neither number alone determines whether chess can eventually be solved.

A complete solution could potentially exploit enormous amounts of mathematical structure rather than explicitly enumerating every game.

At the same time, the existence of a much smaller set of legal positions than possible game sequences does not make the problem trivial. Determining optimal play over that state space remains an extraordinarily difficult computational problem.

The most useful interpretation of the debate is therefore not that one participant produced a single decisive number and the other did not. Rather, the exchange accidentally exposed the distinction between the size of a problem’s search space and the amount of information required to solve it.

That distinction extends well beyond chess.

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